

Unit 4: Project Closing Phase & Agile Frameworks

Experiments Covered in This Unit

Exp 4.1: Project Closure, Final Deliverables and Lessons Learned

Exp 4.2: Agile Scrum Framework - Sprints, Backlog, Velocity

Exp 4.3: Project Management Plan (PMP) - All Subsidiary Plans

Exp 4.4: Kanban, XP, Agile vs Waterfall Comparison

Project Closure Report

Final Delivery Summary

Planned Duration

85 Days

Actual Duration

87 Days (+2)

Planned Budget

INR 10.00 L

Actual Cost

INR 10.55 L

All 14 Deliverables - Completed

- All 5 core modules: AI, Abacus, Vedic Math, Explorer, Quiz
- Firebase Auth + Admin Dashboard fully functional
- GitHub v1.0.0 tagged and production deployed on Vercel
- 96.2% final test pass rate | 0 critical bugs at launch
- All 6 acceptance criteria met | Faculty sign-off obtained

Key Lessons Learned

- AI integration tasks need +20% PERT buffer
- COCOMO misses SaaS API costs - add separate budget line
- Single developer bottleneck - use component libraries aggressively
- Document incrementally during each sprint (not at end)
- Gemini cold-start: always implement warm-up infrastructure

7 Sprints | 197/211 Story Points | 93.4% Velocity

Sprint	Duration	Goal	SP Planned	SP Done	Status
Sprint 1	Day 1-14	Requirements + UI Prototype	28	26	Done
Sprint 2	Day 14-26	Firebase + Authentication	30	29	Done
Sprint 3	Day 26-40	Gemini AI + Story Module	35	27	Done (delayed)
Sprint 4	Day 40-54	Abacus + Vedic Math	38	36	Done
Sprint 5	Day 54-68	Number Explorer + Quiz	32	31	Done
Sprint 6	Day 68-80	Integration + Testing + Mobile	28	28	Done
Sprint 7	Day 80-87	Deployment + Docs + Closure	20	20	Done
TOTAL	87 days	All 6 modules delivered	211	197	v1.0.0 Live

Project Management Plan (PMP)

All Subsidiary Plans at a Glance

PMP Component	Key Content	Reference
Scope Management	WBS, scope statement, acceptance criteria	Unit 1 - Exp 1.1
Schedule Management	PERT/CPM, Gantt, milestone dates	Unit 2 - Exp 2.1 & 2.2
Cost Management	COCOMO estimate, INR 10L baseline, contingency	Unit 1 - Exp 1.3
Quality Management	Jest tests, Lighthouse CI, defect tracking	Unit 3 - Exp 3.4
Resource Management	3-member team, roles, resource levelling	Unit 2 - Exp 2.3
Risk Management	6 risks, RMMM plan, INR 2L reserve	Unit 3 - Exp 3.2 & 3.3
Communication Mgmt	Daily stand-up, sprint review, faculty report	Unit 4 - Exp 4.3
Configuration Mgmt	GitHub Git Flow, CI/CD, SemVer v1.0.0	Unit 3 - Exp 3.3
Procurement Mgmt	Gemini API, Firebase, Vercel, Figma - INR 82K	Unit 1 - Exp 1.3

Scrum vs Kanban vs XP - Applied in This Project

Dimension	Scrum	Kanban	XP
Applied To	Full project lifecycle - feature sprints	Bug management in testing phase	Code quality - TDD, CI, refactoring
Cadence	Fixed 2-week sprints	Continuous pull-based flow	Continuous - no sprint boundary
Key Benefit	Structured delivery + visibility	Reduces context switching	Higher code quality, less tech debt
Best For	Overall project governance	Post-integration defect sprints	AI module where quality is critical
In This Project	7 sprints, 211 SP planned	11 Sprint-2 defects resolved	TDD for abacus + number converter
Primary Tool	GitHub Project Board + Burndown	Kanban board (5 columns, WIP=2)	Jest + GitHub Actions CI + ESLint